Lecture 6

The spectrum and the resolvent

Objectives. *The spectrum of a self-adjoint operator is real and closed. We prove Weyl's criterion and compute spectra by explicit inverses and approximate eigenvectors. By a cited theorem, the spectrum holds the readings. The free resolvent and infinite chain give two applications.*

- Construct a Weyl sequence at a spectral value.
- Distinguish discrete spectrum from essential spectrum.
- Compute the free resolvent kernel and the infinite-chain band.

Lecture 6, first part

Weyl's criterion and the spectrum of a self-adjoint operator

6.1 Resolvent and spectrum

In *Quantum Mechanics I*, Postulate 4.1, the readings of an observable A are its eigenvalues, the numbers λ for which the matrix $A - \lambda$ has no inverse. The position Q on the line has no eigenvalue (Theorem 4.6), yet every interval of positive length is read with positive probability in some state. The test that still works asks for a bounded inverse. Here and below, $A - z$ stands for the operator $A - z\mathbb{1}$ on $D(A)$.

Definition 6.1 (Resolvent and spectrum). Let A be an operator on $\mathcal{H}$. A complex number z lies in the *resolvent set* of A if $A - z$ maps $D(A)$ one-to-one onto $\mathcal{H}$ and its inverse $(A - z)^{-1}$ is bounded. This inverse is the *resolvent* of A at z. The *spectrum* $\sigma(A)$ is the set of complex numbers that do not lie in the resolvent set.

For example, let $A = c\mathbb{1}$ on $\mathcal{H} \neq \{0\}$, with $c \in \mathbb{R}$. If $z \neq c$, then $(A - z)^{-1} = (c - z)^{-1}\mathbb{1}$ and its norm is $1/|c - z|$. At $z = c$ the operator $A - c$ is zero, so $\sigma(A) = \{c\}$. The inverse becomes unbounded as z approaches the one possible reading c.

If A is bounded and $D(A) = \mathcal{H}$, this is the spectrum of *Linear Algebra*, Def. 14.7. Every eigenvalue λ of A lies in $\sigma(A)$, since $A - \lambda$ maps an eigenvector to 0 and is not one-to-one. For z in the resolvent set and every $\varphi \in \mathcal{H}$, the vector $\psi = (A - z)^{-1}\varphi$ is the one solution in $D(A)$ of the equation $(A - z)\psi = \varphi$. An eigenvector shows that a number lies in the spectrum. Without one, approximate eigenvectors do.

Proposition 6.2 (Weyl sequences). *Let A be an operator on $\mathcal{H}$ and $\lambda \in \mathbb{C}$, and suppose that there are unit vectors $\psi_n \in D(A)$ with*

$$\|(A - \lambda)\psi_n\| \to 0.$$

Then $\lambda \in \sigma(A)$, and $\left\|(A - z)^{-1}\right\| \geq 1/|\lambda - z|$ for every z in the resolvent set. Such a sequence

(ψ_n) *is a* Weyl sequence *for* λ.

Proof. Let z lie in the resolvent set. Since $(A - z)\psi_n = (A - \lambda)\psi_n + (\lambda - z)\psi_n$,

$$1 = \left\|(A - z)^{-1}(A - z)\psi_n\right\| \leq \left\|(A - z)^{-1}\right\| \left(\|(A - \lambda)\psi_n\| + |\lambda - z|\right).$$

As $n \to \infty$, this gives $1 \leq \left\|(A - z)^{-1}\right\| |\lambda - z|$. If λ itself lay in the resolvent set, $z = \lambda$ would give $1 \leq 0$. So $\lambda \in \sigma(A)$, and for the other z this is the inequality claimed. ∎

For an eigenvalue, a unit eigenvector, repeated, is a Weyl sequence. For a self-adjoint operator, every point of the spectrum has a Weyl sequence. This is part of the main theorem of this section. We state it now and prove it after the two lemmas that it needs.

Theorem 6.3 (The spectrum of a self-adjoint operator)**.** *Let A be self-adjoint.*

1. *The spectrum $\sigma(A)$ lies in $\mathbb{R}$, and $\left\|(A - z)^{-1}\right\| \leq 1/|\operatorname{Im} z|$ for every non-real z.*
2. Weyl's criterion. *A number λ lies in $\sigma(A)$ if and only if it has a Weyl sequence.*
3. *The spectrum $\sigma(A)$ is closed.*

So λ lies in the spectrum exactly when there are states on which $A - \lambda$ is as small as one likes.

This smallness has a meaning for readings. Let A be self-adjoint, let λ be real, and let $\psi \in D(A)$ be a unit vector with the distribution μ_ψ under E_A. We write a for the variable of integration. By (5.2) and the cited Theorem 5.4,

$$\|(A - \lambda)\psi\|^2 = \|A\psi\|^2 - 2\lambda \langle \psi, A\psi \rangle + \lambda^2 = \int (a - \lambda)^2 \, \mathrm{d}\mu_\psi(a),$$

the mean square distance of the readings from λ. Let $\epsilon > 0$ and $I = (\lambda - \epsilon, \lambda + \epsilon)$. Outside I the integrand is at least ϵ^2, so the probability of a reading outside I satisfies

$$1 - \operatorname{Prob}(A \in I) \leq \frac{\|(A - \lambda)\psi\|^2}{\epsilon^2}.$$

Along a Weyl sequence for a real λ, the right side tends to 0 for every $\epsilon > 0$. So the readings in the states ψ_n *converge in probability* to λ: the probability of a reading outside any fixed interval around λ tends to 0. This says nothing about an eigenvector at λ, and there may be none.

The spectrum also holds the readings, by a theorem that needs the spectral theorem.

Theorem 6.4 (Readings lie in the spectrum · *cited*)**.** *Let A be self-adjoint. A real number λ lies in $\sigma(A)$ if and only if $E_A((\lambda - \epsilon, \lambda + \epsilon)) \neq 0$ for every $\epsilon > 0$, and $E_A(I) = 0$ for every interval I that contains no point of $\sigma(A)$. Moreover, $\left\|(A - z)^{-1}\right\| = 1/\operatorname{dist}(z, \sigma(A))$ for every z in the resolvent set.*

This is Teschl, Thm 3.8 and Cor. 3.9, and Amrein, Example 4.13. By Postulate 5.6, a reading of A in an interval that misses $\sigma(A)$ has probability 0 in every state, and a reading in an open interval around a point of $\sigma(A)$ has positive probability in some state. The spectrum is the continuum's form of the eigenvalues of *Quantum Mechanics I*.

We now prove Theorem 6.3. The proof rests on Lemma 6.6: a lower bound for $A - z$ gives the resolvent. That lemma needs one fact about Hilbert spaces, which *Linear Algebra*, Thm 13.30, cites. We prove it. A subspace M of $\mathcal{H}$ is *closed* if the limit of every convergent sequence in M lies in M.

Lemma 6.5 (Closed subspaces). *Let M be a closed subspace of $\mathcal{H}$. If 0 is the only vector orthogonal to every vector of M, then $M = \mathcal{H}$.*

Proof. Let $\eta \in \mathcal{H}$, let d be the infimum of $\|\eta - \mu\|$ over $\mu \in M$, and choose $\mu_n \in M$ with $\|\eta - \mu_n\|^2 < d^2 + 1/n$. The parallelogram law for the vectors $\eta - \mu_n$ and $\eta - \mu_k$ gives

$$\|\mu_n - \mu_k\|^2 = 2\,\|\eta - \mu_n\|^2 + 2\,\|\eta - \mu_k\|^2 - 4\left\|\eta - \frac{\mu_n + \mu_k}{2}\right\|^2 < \frac{2}{n} + \frac{2}{k},$$

since $(\mu_n+\mu_k)/2$ lies in M and so has distance at least d from η. So (μ_n) is a Cauchy sequence. Since M is closed, its limit μ lies in M, and $\|\eta - \mu\| = d$, since $\|\eta - \mu_n\| \to d$. Let $\nu \in M$ and $w = \langle \eta - \mu, \nu \rangle$. For every complex number t, the vector $\mu + t\nu$ lies in M, so

$$d^2 \le \|\eta - \mu - t\nu\|^2 = d^2 - 2\,\mathrm{Re}(t\,w) + |t|^2\,\|\nu\|^2\,.$$

With $t = \epsilon\overline{w}$ for small $\epsilon > 0$, this reads $\epsilon\,|w|^2\,(\epsilon\,\|\nu\|^2 - 2) \ge 0$. The bracket is negative for small ϵ, so $w = 0$. Hence $\eta - \mu$ is orthogonal to every vector of M. By the hypothesis, $\eta - \mu = 0$, so $\eta = \mu$ lies in M. ∎

Let A be symmetric, let $z = a + ib$ with a and b real, and let $\psi \in D(A)$. Then $(A - z)\psi = (A - a)\psi - ib\psi$. In the square of the norm of this sum, the cross term

$$2\,\mathrm{Re}\,\langle (A - a)\psi, -ib\psi \rangle = 2\,\mathrm{Re}\big(-ib\,\langle (A - a)\psi, \psi \rangle\big)$$

vanishes, since $\langle (A - a)\psi, \psi \rangle$ is real (§ 0.2, self-check). So

$$\|(A - z)\psi\|^2 = \|(A - a)\psi\|^2 + b^2\,\|\psi\|^2\,. \tag{6.1}$$

Replacing b by $-b$ shows that $\|(A - z)\psi\| = \|(A - \overline{z})\psi\|$.

Lemma 6.6 (A lower bound gives the resolvent). *Let A be self-adjoint, let $z \in \mathbb{C}$ and $c > 0$, and suppose that*

$$\|(A - z)\psi\| \ge c\,\|\psi\| \quad \text{for every } \psi \in D(A).$$

Then z lies in the resolvent set of A, and $\left\|(A - z)^{-1}\right\| \le 1/c$.

Proof. First, $A - z$ is one-to-one, since $(A - z)\psi = 0$ gives $c\,\|\psi\| \le 0$.

Next, the range of $A - z$ is closed. Let $(A - z)\psi_n \to \eta$. By the bound, $\|\psi_n - \psi_k\| \le \|(A - z)\psi_n - (A - z)\psi_k\|\,/c$, so (ψ_n) is a Cauchy sequence, with a limit ψ. For every $\varphi \in D(A)$, the symmetry of A gives $\langle (A - \overline{z})\varphi, \psi_n \rangle = \langle \varphi, (A - z)\psi_n \rangle$, and in the limit $\langle A\varphi, \psi \rangle - z\,\langle \varphi, \psi \rangle = \langle \varphi, \eta \rangle$. Taking complex conjugates,

$$\langle \psi, A\varphi \rangle = \langle \eta + z\psi, \varphi \rangle \quad \text{for every } \varphi \in D(A).$$

So $\psi \in D(A^\dagger)$, with $A^\dagger\psi = \eta + z\psi$ (Definition 0.2). Since $A^\dagger = A$, the vector ψ lies in $D(A)$, and $(A - z)\psi = \eta$.

Then, only 0 is orthogonal to the range. Let $\langle \varphi, (A - z)\psi \rangle = 0$, that is, $\langle \varphi, A\psi \rangle = \langle \overline{z}\varphi, \psi \rangle$, for every $\psi \in D(A)$. So, by Definition 0.2, $\varphi \in D(A^\dagger) = D(A)$, with $A\varphi = \overline{z}\varphi$. By the bound and (6.1), $c\,\|\varphi\| \le \|(A - z)\varphi\| = \|(A - \overline{z})\varphi\| = 0$, so $\varphi = 0$. Since the range of $A - z$ is closed, Lemma 6.5 shows that it is $\mathcal{H}$.

Finally, if $(A - z)\psi = \eta$, then $\|\psi\| \le \|\eta\|\,/c$. So the inverse is bounded, with $\left\|(A - z)^{-1}\right\| \le 1/c$. ∎

Proof of Theorem 6.3.

1. For non-real z, (6.1) gives $\|(A-z)\psi\| \geq |\operatorname{Im} z|\,\|\psi\|$, and Lemma 6.6 applies with $c = |\operatorname{Im} z|$.
2. By Proposition 6.2, a Weyl sequence for λ puts λ in $\sigma(A)$. Conversely, suppose that λ has no Weyl sequence. Then there is $c > 0$ with $\|(A-\lambda)\psi\| \geq c\,\|\psi\|$ for every $\psi \in D(A)$, since otherwise for each n some unit vector ψ_n would have $\|(A-\lambda)\psi_n\| < 1/n$. By Lemma 6.6, λ lies in the resolvent set.
3. Let $\lambda_k \in \sigma(A)$ converge to λ. By Weyl's criterion there are unit vectors $\psi_k \in D(A)$ with $\|(A-\lambda_k)\psi_k\| < 1/k$. Then $\|(A-\lambda)\psi_k\| < 1/k + |\lambda_k - \lambda| \to 0$, so $\lambda \in \sigma(A)$. ∎

Self-adjointness can be tested without computing $A^\dagger$, by solving two equations.

Proposition 6.7 (A test for self-adjointness)**.** *Let A be symmetric, and suppose that for one non-real z both $A - z$ and $A - \overline{z}$ map $D(A)$ onto $\mathcal{H}$. Then A is self-adjoint.*

Proof. Let $\varphi \in D(A^\dagger)$, with $\chi = A^\dagger\varphi$. Since $A - \overline{z}$ maps onto $\mathcal{H}$, there is $\varphi_0 \in D(A)$ with $(A-\overline{z})\varphi_0 = \chi - \overline{z}\varphi$. For every $\psi \in D(A)$, the definition of χ (Definition 0.2) and the symmetry of A give

$$\langle \varphi, (A-z)\psi\rangle = \langle \chi - \overline{z}\varphi, \psi\rangle = \langle (A-\overline{z})\varphi_0, \psi\rangle = \langle \varphi_0, (A-z)\psi\rangle\,.$$

So $\varphi - \varphi_0$ is orthogonal to the range of $A - z$, which is $\mathcal{H}$. Hence $\varphi - \varphi_0 = 0$, and $\varphi = \varphi_0 \in D(A)$. Hence $D(A^\dagger) \subseteq D(A)$, and with the symmetry, $A^\dagger = A$. ∎

By the first claim of Theorem 6.3, a self-adjoint A passes the test for every non-real z, so the test is also necessary. Lecture 7 uses it with $z = \mathrm{i}/\tau$, for a time τ, to prove Stone's theorem.

So the readings of a self-adjoint observable lie in its spectrum, by the cited Theorem 6.4, and the spectrum is found without a diagonalisation: a real number lies in it exactly when it has a Weyl sequence.

Self-check 6.1

1. Take the packet (4.3) with centre q and width s. Bound the probability of a position reading outside $(q-\epsilon, q+\epsilon)$. For which widths s_n do these packets form a Weyl sequence for Q at q? What do their position readings then do?

Lecture 6, second part

The spectra of the box, the line and the infinite chain

6.2 Spectra of the operators of this course

The first part proved that a real number lies in the spectrum of a self-adjoint operator exactly when it has a Weyl sequence. In this part we find the spectra of the operators of this course, sort their points into kinds, and compute the free resolvent and the band of the infinite chain. For the operators that Lectures 1, 3 and 4 diagonalised, the spectrum is found by hand, with an explicit inverse off the spectrum and Weyl sequences on it.

Proposition 6.8 (Spectra of diagonal and multiplication operators)**.**

1. Diagonal. *Let B be the operator made from an orthonormal basis (e_k) and real numbers λ_k, as in Proposition 5.5. Then $\sigma(B)$ is the closure of the set of the λ_k.*
2. Multiplication. *Let g be a real continuous function on $\mathbb{R}$, with M_g as in Lemma 0.7, let U be unitary, and let $B = U^{-1}M_gU$. Then $\sigma(B)$ is the closure of the set of values of*

g.

In both cases $\left\|(B-z)^{-1}\right\| = 1/\operatorname{dist}(z,\sigma(B))$ *for every* z *in the resolvent set.*

Proof.

1. Let Λ be the closure of the set of the λ_k, and let $z \notin \Lambda$, so that $d = \operatorname{dist}(z,\Lambda) > 0$. Write $c_k = \langle e_k, \psi\rangle$ and put
$$R\psi = \sum_k \frac{c_k}{\lambda_k - z}\, e_k.$$
Since $|\lambda_k - z| \geq d$, Parseval's identity gives $\|R\psi\| \leq \|\psi\|/d$. Since $|\lambda_k| \leq |\lambda_k - z| + |z|$, the coefficients $\lambda_k c_k/(\lambda_k - z)$ have moduli at most $(1 + |z|/d)\,|c_k|$, so $R\psi \in D(B)$ and $(B-z)R\psi = \psi$. For $\psi \in D(B)$, $R(B-z)\psi = \psi$ term by term. So z lies in the resolvent set, and $\left\|(B-z)^{-1}\right\| = \|R\| \leq 1/d$. Conversely, let $\lambda \in \Lambda$, and choose indices k_n with $\left|\lambda_{k_n} - \lambda\right| < 1/n$. Then $\left\|(B-\lambda)e_{k_n}\right\| < 1/n$, so (e_{k_n}) is a Weyl sequence, and $\lambda \in \sigma(B)$ by Proposition 6.2. Hence $\sigma(B) = \Lambda$, $\operatorname{dist}(z,\sigma(B)) = d$, and Proposition 6.2 gives $\left\|(B-z)^{-1}\right\| \geq 1/d$.
2. First let $U = \mathbb{1}$. Let Λ be the closure of the set of values of g, let $z \notin \Lambda$, and let $d = \operatorname{dist}(z,\Lambda) > 0$. As in the diagonal case, multiplication by the continuous function $1/(g-z)$, which is bounded by $1/d$, is the inverse of $M_g - z$, since $|g/(g-z)| \leq 1 + |z|/d$. So $\left\|(M_g - z)^{-1}\right\| \leq 1/d$. Conversely, let $\lambda \in \Lambda$, and choose points x_n with $|g(x_n) - \lambda| < 1/n$. Since g is continuous, there is $\delta_n > 0$ with $|g(x) - g(x_n)| < 1/n$ for $|x - x_n| \leq \delta_n$. The unit vectors $\psi_n = (2\delta_n)^{-1/2}\, 1_{[x_n - \delta_n,\, x_n + \delta_n]}$ then have
$$\left\|(M_g - \lambda)\psi_n\right\| \leq \max_{|x - x_n| \leq \delta_n} |g(x) - \lambda| < \frac{2}{n},$$
so they form a Weyl sequence. As in the diagonal case, $\sigma(M_g) = \Lambda$ and $\left\|(M_g - z)^{-1}\right\| = 1/d$. Finally, $B - z = U^{-1}(M_g - z)U$. So z lies in the resolvent set of B exactly when it lies in that of M_g, and then $(B-z)^{-1} = U^{-1}(M_g - z)^{-1}U$ has the same norm. The vectors $U^{-1}\psi_n$ form a Weyl sequence for B. ∎

By the diagonal case and Theorem 1.8, the box Hamiltonian has $\sigma(H) = \{E_n : n \geq 1\}$, since the levels $E_n = n^2 E_1$ have no finite limit point. In the same way, the momentum on the circle has the spectrum $\{2\pi\hbar k/L : k \in \mathbb{Z}\}$ (Proposition 3.2). By the multiplication case with $g(x) = x$, $\sigma(Q) = \mathbb{R}$. With the Fourier transform as U (Theorem 4.6), $\sigma(P) = \mathbb{R}$, and $\sigma(H_0) = [0,\infty)$, the set of values of $p^2/2m$. Every real number lies in the spectrum of Q and of P, though none is an eigenvalue. By Theorem 6.4, the free energy reads in $(-\infty, 0)$ with probability 0.

Physics texts call the plane wave $e^{\mathrm{i}p_0 x/\hbar}$ an eigenvector of P with the eigenvalue p_0. As § 4.2 showed, it is not a state. What is true is that longer and longer pieces of it form a Weyl sequence. Let f be a real smooth function that vanishes outside $[-1,1]$, with $\|f\| = 1$, and put
$$\psi_n(x) = n^{-1/2} f(x/n)\, e^{\mathrm{i}p_0 x/\hbar}.$$
Substituting $x = ny$ shows that $\|\psi_n\| = \|f\| = 1$. Each ψ_n lies in $\mathcal{S}(\mathbb{R})$, so by Theorem 4.6
$$(P - p_0)\psi_n = -\mathrm{i}\hbar\psi_n' - p_0\psi_n = -\mathrm{i}\hbar\, n^{-3/2} f'(x/n)\, e^{\mathrm{i}p_0 x/\hbar},$$
and the same substitution gives
$$\|(P - p_0)\psi_n\| = \frac{\hbar}{n}\, \|f'\| \longrightarrow 0.$$

Two derivatives give a Weyl sequence for H_0 at $p_0^2/2m$ in the same way. The terms with $f(x/n)$ cancel, and

$$\Bigl(H_0 - \frac{p_0^2}{2m}\Bigr)\psi_n = -\frac{\hbar^2}{2m}\, n^{-1/2}\Bigl(\frac{2\mathrm{i}p_0}{\hbar n}\, f'(x/n) + \frac{1}{n^2}\, f''(x/n)\Bigr)e^{\mathrm{i}p_0 x/\hbar},$$

and by the same substitution its norm is at most $\hbar\,|p_0|\,\|f'\|/mn + \hbar^2\,\|f''\|/2mn^2$, which tends to 0.

Figure 6.1 draws ψ_n for $n = 1$, 2 and 4, at $\hbar = 1$ and $p_0 = 2\pi$. The figure uses $f(x) = Ce^{-1/(1-x^2)}$ for $|x| < 1$ and $f(x) = 0$ otherwise, with $C > 0$ chosen so that $\|f\| = 1$. As n grows, the wavelength $2\pi\hbar/p_0$, here 1, stays the same, the support $[-n, n]$ grows, and the height falls like $n^{-1/2}$.

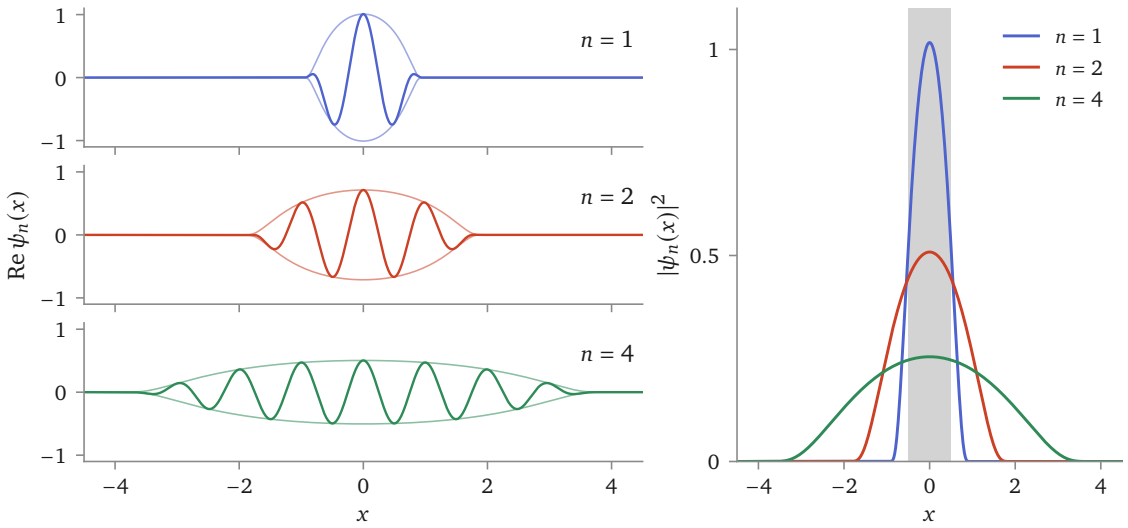


Figure 6.1: The cut-off plane waves ψ_n for $n = 1$, 2 and 4, one colour for each n, at $\hbar = 1$ and $p_0 = 2\pi$. The rows on the left draw the real part of ψ_n between the curves $\pm|\psi_n|$. The panel on the right draws the densities $|\psi_n|^2$, each of area 1. The area over the grey interval $J = [-\frac{1}{2}, \frac{1}{2}]$ is $\|1_J\psi_n\|^2$, and it falls as n grows.

The unnormalised packets approach a multiple of the plane wave:

$$n^{1/2}\psi_n(x) = f(x/n)e^{\mathrm{i}p_0 x/\hbar} \longrightarrow f(0)e^{\mathrm{i}p_0 x/\hbar}.$$

The convergence is uniform on every bounded interval, since

$$|f(x/n) - f(0)| \le \max|f'|\;|x|/n.$$

The states ψ_n themselves approach no vector. Since $|\psi_n|^2 \le \max|f|^2/n$, for every bounded interval J of length $|J|$

$$\|1_J\psi_n\|^2 \le \frac{|J|\max|f|^2}{n} \longrightarrow 0,$$

so the probability of finding the particle in any fixed interval tends to 0. The sequence has no limit in $L^2(\mathbb{R})$. A limit ψ would have $\|1_J\psi\| = \lim\|1_J\psi_n\| = 0$ for every J, so $\psi = 0$, against $\|\psi\| = 1$. The sequence becomes orthogonal in the limit to every fixed vector, that is, $\langle\chi, \psi_n\rangle \to 0$ for every $\chi \in L^2(\mathbb{R})$. Indeed, for every bounded interval J,

$$|\langle\chi, \psi_n\rangle| \le \|\chi\|\;\|1_J\psi_n\| + \|(1 - 1_J)\chi\|\,.$$

The second term is small for large J, and the first then tends to zero.

Self-check 6.2

1. Show that $\psi_n = \sqrt{n/2}\, 1_{[\lambda-1/n,\, \lambda+1/n]}$ is a Weyl sequence for the position Q at the real number λ.

6.3 Kinds of spectrum

Let A be self-adjoint. Its eigenvalues form the *point spectrum*, and the other points of $\sigma(A)$ form the *continuous spectrum*. These are the names of *Linear Algebra*, Def. 14.7, since for a self-adjoint A and a real λ that is not an eigenvalue, the range of $A - \lambda$ is dense. Indeed, as in the proof of Lemma 6.6, a vector φ orthogonal to the range lies in $D(A)$ and has $A\varphi = \lambda\varphi$, so $\varphi = 0$. Then Lemma 6.5, applied to the closure of the range, shows that the range is dense. The spectra of Q, P and H_0 are continuous spectrum only, and that of the box Hamiltonian is point spectrum only. Physics needs a second division, by isolation.

Definition 6.9 (Discrete and essential spectrum)**.** Let A be self-adjoint. A point $\lambda \in \sigma(A)$ lies in the *discrete spectrum* $\sigma_{\mathrm{d}}(A)$ if it is an eigenvalue with an eigenspace of finite dimension, and some interval $(\lambda - \epsilon, \lambda + \epsilon)$ contains no other point of $\sigma(A)$. The *essential spectrum* $\sigma_{\mathrm{ess}}(A)$ is the rest of $\sigma(A)$.

So a point of the essential spectrum is a point of the continuous spectrum, an eigenvalue of infinite multiplicity, or a limit of other points of the spectrum. The box Hamiltonian has $\sigma_{\mathrm{d}}(H) = \sigma(H)$, since each E_n is simple and isolated. The position, the momentum and the free energy on the line have $\sigma_{\mathrm{ess}} = \sigma$, since they have no eigenvalues.

Theorem 6.10 (Weyl's criterion for the essential spectrum · *cited*)**.** *Let A be self-adjoint and λ real. Then $\lambda \in \sigma_{\mathrm{ess}}(A)$ if and only if λ has a Weyl sequence of orthonormal vectors.*

This is Teschl, Lemma 6.17, Amrein, Prop. 4.20, and Prugovečki, Thm V.5.11. An isolated eigenvalue of finite multiplicity has a Weyl sequence, its eigenvector repeated, but, by the theorem, no orthonormal one. An interval of positive length inside $\sigma(A)$ lies in $\sigma_{\mathrm{ess}}(A)$ without the theorem, since none of its points is isolated. Lecture 10 shows in this way, with Weyl sequences sent to infinity, that a bounded potential that vanishes outside a bounded interval leaves $[0, \infty)$ in the essential spectrum.

Linear Algebra, Lecture 14, warns that a continuous spectrum is not a continuous distribution of readings. A diagonal operator can have continuous spectrum, though all its distributions are point weights, as the self-check below shows. Lecture 12 sorts states by their distributions, not by the continuous spectrum.

An eigenvalue can also lie inside an interval of essential spectrum. Put two impenetrable walls on the line, at 0 and at L. The state space splits into the functions on the three intervals, and the energy is then the direct sum of three Dirichlet operators, on $(-\infty, 0)$, on $(0, L)$ and on (L, ∞). The operator acts by $-(\hbar^2/2m)\psi''$ on each interval separately. On each interval, ψ and ψ' are absolutely continuous on every bounded subinterval, ψ'' is square-integrable, and ψ vanishes at each finite endpoint. No condition joins the derivatives across a wall. This operator is self-adjoint (Albeverio et al., p. 141, cited). Each box vector u_n, extended by zero outside $(0, L)$, is an eigenvector with the eigenvalue E_n. On the two half-lines the particle is free, and by Albeverio et al., p. 141 and Thm II.2.1.3, cited, the energy has the essential spectrum $[0, \infty)$. So each E_n is an eigenvalue with points of the spectrum on both sides, *embedded* in the essential spectrum.

Self-check 6.3

1. Let B be the diagonal operator with $\lambda_k = 1/k$, for $k \geq 1$. Find $\sigma(B)$, $\sigma_{\mathrm{d}}(B)$ and $\sigma_{\mathrm{ess}}(B)$. Is 0 an eigenvalue?

Figure 6.2 collects the spectra of this section. The two divisions differ. An eigenvalue can be discrete or essential. A point of the continuous spectrum is always essential, since a discrete point is an eigenvalue. The shape of a mark tells the first division, and its colour tells the second.

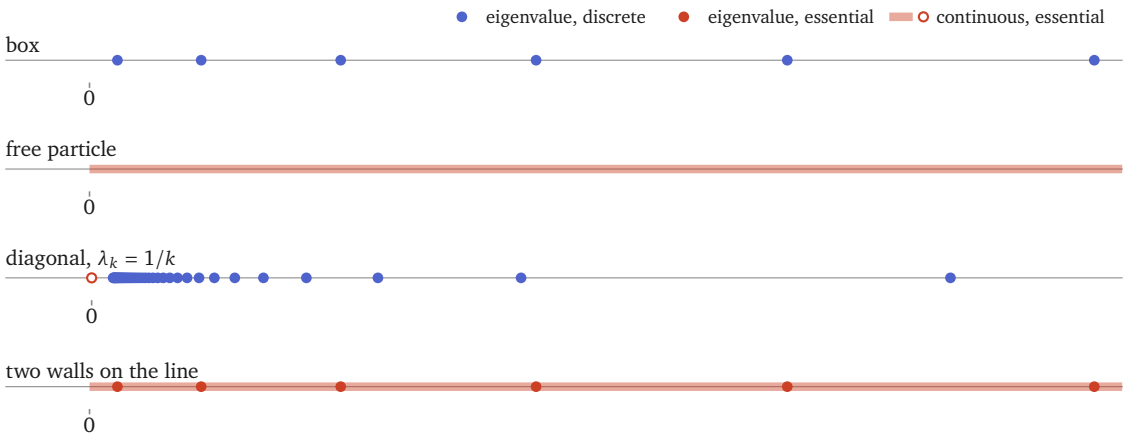


Figure 6.2: The spectra of this section, one per row, as sets of real numbers. The diagonal row has its own scale, and the other rows share one. On the diagonal row only finitely many dots are drawn, and the circle at 0 is continuous spectrum, a limit of eigenvalues that is not itself an eigenvalue. On the last row each box level is an eigenvalue inside the essential spectrum, with continuous spectrum on both sides.

6.4 The free resolvent

On the line the resolvent of the free energy is an integral, with an explicit kernel. The decay rate is written η, since κ is the hopping of *Quantum Mechanics I*.

Proposition 6.11 (The free resolvent). *Let $\eta > 0$ and $z = -\hbar^2\eta^2/2m$. Then z lies in the resolvent set of H_0, $\|(H_0 - z)^{-1}\| = 1/|z|$, and for every smooth φ that vanishes outside a bounded interval*

$$\big((H_0 - z)^{-1}\varphi\big)(x) = \int_{-\infty}^{\infty} G(x-y)\,\varphi(y)\,\mathrm{d}y, \qquad G(x) = \frac{m}{\hbar^2\eta}\,e^{-\eta|x|}.$$

Proof. By Proposition 6.8, $\sigma(H_0) = [0, \infty)$, whose distance from z is $|z|$. This gives the first two claims. Let u be the integral, split at $y = x$:

$$u(x) = \frac{m}{\hbar^2\eta}\Big(e^{-\eta x}\int_{-\infty}^{x} e^{\eta y}\varphi(y)\,\mathrm{d}y + e^{\eta x}\int_{x}^{\infty} e^{-\eta y}\varphi(y)\,\mathrm{d}y\Big).$$

By the fundamental theorem of calculus, the two terms $\varphi(x)$ that come from the limits cancel in u':

$$u'(x) = \frac{m}{\hbar^2}\Big(-e^{-\eta x}\int_{-\infty}^{x} e^{\eta y}\varphi(y)\,\mathrm{d}y + e^{\eta x}\int_{x}^{\infty} e^{-\eta y}\varphi(y)\,\mathrm{d}y\Big).$$

In u'' they add up:

$$u''(x) = \frac{m}{\hbar^2}\Big(\eta e^{-\eta x}\int_{-\infty}^{x} e^{\eta y}\varphi(y)\,dy + \eta e^{\eta x}\int_{x}^{\infty} e^{-\eta y}\varphi(y)\,dy - 2\varphi(x)\Big) = \eta^2 u(x) - \frac{2m}{\hbar^2}\varphi(x).$$

So

$$-\frac{\hbar^2}{2m}u'' - zu = -\frac{\hbar^2}{2m}u'' + \frac{\hbar^2\eta^2}{2m}u = \varphi.$$

By the equation $u'' = \eta^2 u - (2m/\hbar^2)\varphi$ and induction, u has derivatives of every order. If φ vanishes outside $[-R, R]$, then u is a multiple of $e^{-\eta x}$ for $x > R$ and of $e^{\eta x}$ for $x < -R$. So every $x^a u^{(b)}$ is bounded, and $u \in \mathcal{S}(\mathbb{R})$. By Theorem 4.6, $H_0 u = -(\hbar^2/2m)u''$, so $(H_0 - z)u = \varphi$. Since $H_0 - z$ is one-to-one, $u = (H_0 - z)^{-1}\varphi$. ∎

By the standing fact S4, these φ are dense, so the kernel G fixes the bounded operator $(H_0 - z)^{-1}$. The kernel is the response at x to a source at y. It decays over the length $1/\eta = \hbar/\sqrt{2m\,|z|}$, and its integral $\int G = 2m/\hbar^2\eta^2$ equals the norm $1/|z|$ of the resolvent. As z rises to the edge 0 of $\sigma(H_0)$, the decay length and the height $G(0) = m/\hbar^2\eta$ grow without bound. Lecture 11 uses G to find the level of a narrow well.

Self-check 6.4

1. For an electron at $z = -1\,\text{eV}$, how long is the decay length $1/\eta$ of the kernel G? Use $\hbar c = 197.3\,\text{eV nm}$ and $mc^2 = 511\,\text{keV}$.

6.5 The infinite chain

Quantum Mechanics I, Thm 8.1, found the levels $\varepsilon_k = c - 2\kappa\cos(2\pi k/N)$ of a ring of N sites, and said after Example 8.4 that the continuum is the limit $N \to \infty$. Here is that limit, at a fixed spacing. On $\ell^2(\mathbb{Z})$, with the basis vectors e_j for $j \in \mathbb{Z}$, let T be the shift $Te_j = e_{j+1}$, so that $(T\psi)_j = \psi_{j-1}$. For real c and $\kappa > 0$, the Hamiltonian $H = c\mathbb{1} - \kappa(T + T^\dagger)$ of *Quantum Mechanics I* acts by

$$(H\psi)_j = c\psi_j - \kappa(\psi_{j+1} + \psi_{j-1}).$$

The shift keeps norms, and $T^\dagger$ is the inverse shift. So H is bounded, defined on every vector, and self-adjoint.

Proposition 6.12 (The infinite chain). *The spectrum of the chain Hamiltonian is the band $\sigma(H) = [c - 2\kappa,\ c + 2\kappa]$, and H has no eigenvalue. So $\sigma(H)$ is continuous spectrum and essential spectrum.*

Proof. First, a unitary carries H to a multiplication operator. By the standing fact S2, with $L = 2\pi$ and the interval shifted by π, the functions $g_j(\xi) = e^{-ij\xi}/\sqrt{2\pi}$, for $j \in \mathbb{Z}$, form an orthonormal basis of $L^2(-\pi, \pi)$. So $V\psi = \sum_j \psi_j g_j$ defines a unitary $V : \ell^2(\mathbb{Z}) \to L^2(-\pi, \pi)$. Since $g_{j+1} = e^{-i\xi}g_j$, $V(T\psi) = \sum_j \psi_j g_{j+1} = e^{-i\xi}V\psi$, and in the same way $V(T^\dagger\psi) = e^{i\xi}V\psi$. So

$$V(H\psi) = \varepsilon\, V\psi, \qquad \varepsilon(\xi) = c - 2\kappa\cos\xi.$$

Next, let λ lie outside the band, at the distance $d > 0$ from it. Multiplication by $1/(\varepsilon - \lambda)$, bounded by $1/d$, inverts multiplication by $\varepsilon - \lambda$, so $(H - \lambda)^{-1}$ is V^{-1} times this multiplication times V, and $\lambda \notin \sigma(H)$.

Then, let λ lie in the band. The function ε is continuous on $[-\pi, \pi]$, so the Weyl sequences of the proof of Proposition 6.8, with $(-\pi, \pi)$ in place of $\mathbb{R}$ and each δ_n so small that the

bump lies in $(-\pi, \pi)$, show that λ lies in the spectrum of multiplication by ε, and V^{-1} carries them to Weyl sequences for H.

Finally, if $H\psi = \lambda\psi$, then $(\varepsilon - \lambda)V\psi = 0$. The function $\varepsilon - \lambda$ vanishes at no more than two points of $(-\pi, \pi]$, and the last step of the proof of Lemma 0.7, on $(-\pi, \pi)$ instead of $\mathbb{R}$, gives $V\psi = 0$. So $\psi = 0$, and H has no eigenvalue. Hence $\sigma(H)$ is continuous spectrum, and $\sigma_{\mathrm{d}}(H)$ is empty. ∎

The N levels of the ring lie in the band, and as N grows they fill it. Figure 6.3 draws them for $N = 6$, 12 and 24. The infinite chain keeps the whole band as its spectrum and loses every eigenvector, so a bounded operator can have no eigenvalue. Like the plane wave, the *Bloch wave* $(e^{\mathrm{i}\xi j})_j$ satisfies the eigenvalue equation for $\varepsilon(\xi)$ but is not a vector of the space, and cut off it gives a Weyl sequence (self-check below).

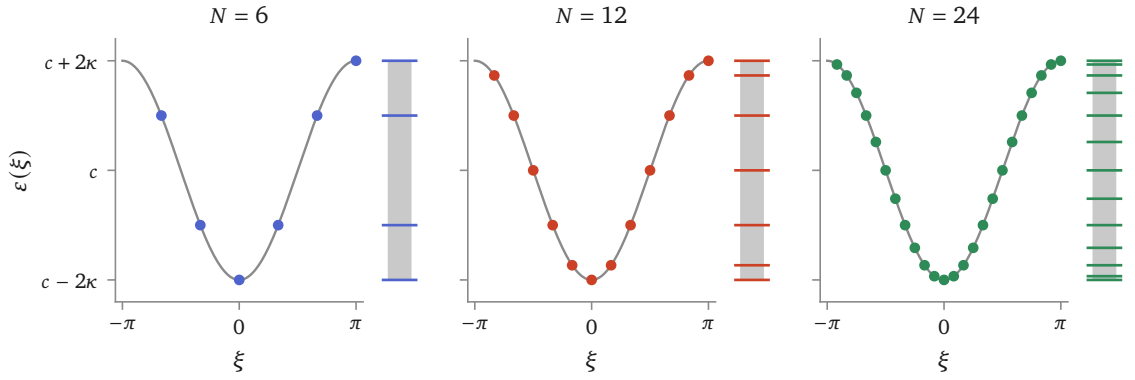


Figure 6.3: The function $\varepsilon(\xi) = c - 2\kappa\cos\xi$ on $[-\pi, \pi]$ (grey), with $\kappa > 0$, and the levels $\varepsilon(2\pi k/N)$ of a ring of N sites as dots at the points $\xi = 2\pi k/N$ in $(-\pi, \pi]$. Beside each graph, the grey bar is the band $[c - 2\kappa, c + 2\kappa]$, and each line is one level. The levels at ξ and $-\xi$ coincide, so a line can stand for two of the N levels.

Self-check 6.5

1. Let $n \geq 2$, and let $\psi_j = e^{\mathrm{i}\xi j}/\sqrt{n}$ for $1 \leq j \leq n$ and $\psi_j = 0$ otherwise. Show that $(H - \varepsilon(\xi))\psi$ has four nonzero entries, each of modulus $\kappa/\sqrt{n}$. For which ξ in $[0, \pi]$ is $\varepsilon(\xi) = c$?

The physics

In 1885 Balmer fitted the four visible lines of hydrogen, at 656, 486, 434 and 410 nm, by the wavelengths $Bj^2/(j^2 - 4)$ for $j = 3, 4, 5, 6$, with $B = 364.56$ nm. As j grows, the lines crowd towards the wavelength B. Each line is a jump between two levels of the hydrogen atom's energy. The levels accumulate at the energy at which the electron is freed, so this point lies in the essential spectrum, and above it lies a continuum of essential spectrum. Light of a wavelength shorter than B can free an electron from the second level, and the spectra of stars such as Vega, which absorb in this way, show a step at about 365 nm. Lecture 14 finds the continuum and Lecture 15 the levels.

The spectrum and the resolvent

Observables: the spectrum of a self-adjoint A is real, and a real λ lies in it exactly when λ has a Weyl sequence (Theorem 6.3). The spectra of Q and P are $\mathbb{R}$, that of H_0 is $[0,\infty)$, that of the box Hamiltonian is the set of the E_n (Proposition 6.8), and that of the infinite chain is a band with no eigenvalue (Proposition 6.12). *Measurement:* by a cited theorem, a reading of A in an interval that contains no point of $\sigma(A)$ has probability 0 (Theorem 6.4).

Summary.

- *Resolvent and spectrum* (Definition 6.1). A Weyl sequence for λ puts λ in the spectrum (Proposition 6.2). The spectrum of a self-adjoint operator is real and closed, and it consists of the points with Weyl sequences (Theorem 6.3); by a cited theorem, it holds the readings (Theorem 6.4). Along a Weyl sequence for a real λ, the readings converge in probability to λ. A symmetric operator with $A - z$ and $A - \overline{z}$ onto $\mathcal{H}$ for one non-real z is self-adjoint (Proposition 6.7).
- *Spectra by hand*: a diagonal operator, and multiplication by a real continuous function, have the closure of their values as spectrum (Proposition 6.8).
- *Kinds of spectrum*: point and continuous, discrete and essential (Definition 6.9); a point is essential exactly when it has an orthonormal Weyl sequence (Theorem 6.10, cited), and an interval of positive length inside the spectrum is essential.
- *The free resolvent* has the kernel $(m/\hbar^2\eta)e^{-\eta|x-y|}$ at $z = -\hbar^2\eta^2/2m$ (Proposition 6.11), and *the infinite chain* has the band $[c-2\kappa, c+2\kappa]$ and no eigenvalue (Proposition 6.12).

Lecture 7 uses the test of Proposition 6.7 to prove Stone's theorem: the generator of a unitary evolution is self-adjoint.

Exercises. Exercise Sheet 6.

Self-check answers.

6.1 1. Since the mean of Q is q, $\|(Q-q)\psi\|^2 = (\Delta Q)^2 = s^2$, so the probability is at most s^2/ϵ^2. The packets with widths s_n form a Weyl sequence exactly when $s_n \to 0$. Their position readings then converge in probability to q, though q is not an eigenvalue of Q.

6.2 1. Each ψ_n is a unit vector in $D(Q)$, and $\|(Q-\lambda)\psi_n\|^2 = \frac{n}{2}\int_{\lambda-1/n}^{\lambda+1/n}(x-\lambda)^2\,\mathrm{d}x = \frac{1}{3n^2} \to 0$.

6.3 1. $\sigma(B) = \{0\} \cup \{1/k : k \geq 1\}$, the closure of the values. Each $1/k$ is simple and isolated, so $\sigma_{\mathrm{d}}(B) = \{1/k : k \geq 1\}$, and $\sigma_{\mathrm{ess}}(B) = \{0\}$, a limit of eigenvalues. It is not an eigenvalue: $B\psi = 0$ gives $c_k/k = 0$ for every k.

6.4 1. $1/\eta = \hbar/\sqrt{2m\,|z|} = \hbar c/\sqrt{2mc^2\,|z|} = 197.3/\sqrt{1.022\times10^6}\,\mathrm{nm} \approx 0.195\,\mathrm{nm}$.

6.5 1. For $2 \leq j \leq n-1$, $(H\psi)_j = \varepsilon(\xi)\psi_j$, since $e^{\mathrm{i}\xi(j+1)} + e^{\mathrm{i}\xi(j-1)} = 2\cos\xi\, e^{\mathrm{i}\xi j}$. At $j=1$ the term $\psi_0 = 0$ is missing, and the entry is $\kappa/\sqrt{n}$; at $j=n$ it is $\kappa e^{\mathrm{i}\xi(n+1)}/\sqrt{n}$; at $j=0$ and $j=n+1$ the entries are $-\kappa\psi_1$ and $-\kappa\psi_n$. So the residual is $2\kappa/\sqrt{n}$. Next, $\varepsilon(\xi) = c$ exactly for $\xi = \pi/2$.

References

Other books prove some results of this lecture. Weyl's criterion of Theorem 6.3 is in Teschl, Thm 2.18 and Lemma 2.16, and in Amrein, Prop. 4.20. The test of Proposition 6.7 is Teschl, Lemma 2.3. The description of the essential spectrum after Definition 6.9 is in Amrein,

p. 164. The cut-off plane waves of § 6.2, which become orthogonal in the limit to every fixed vector, are in Gustafson and Sigal, pp. 395–396.

Teschl, *Mathematical Methods in Quantum Mechanics*, Lemma 2.3 (p. 63), Lemma 2.16 (p. 76), Thm 2.18 (p. 77), Thm 3.8 and Cor. 3.9 (p. 97), Lemma 6.17 (p. 145). Amrein, *Hilbert Space Methods in Quantum Mechanics*, Example 4.13 (p. 155), p. 164, Prop. 4.20 (p. 164). Prugovečki, *Quantum Mechanics in Hilbert Space*, Thm V.5.11 (p. 537). Gustafson and Sigal, *Mathematical Concepts of Quantum Mechanics*, pp. 395–396. Albeverio, Gesztesy, Høegh-Krohn and Holden, *Solvable Models in Quantum Mechanics*, p. 141 and Thm II.2.1.3 (p. 142). *Linear Algebra for Physics*, Thm 13.30, Def. 14.7. *Quantum Mechanics I*, Postulate 4.1, Thm 8.1 and Example 8.4. Balmer, *Ann. Phys. Chem.* 25, 80 (1885). Bohlin and Gilliland, *Astron. J.* 127, 3508 (2004).